DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Web Application Development Laboratory — Academic Year 2026

UNIT 4 PRACTICAL LABORATORY RECORD SCENARIO 4: HOTEL ROOM BOOKING WEB APPLICATION

Comprehensive Technical Implementation & Verification Dossier

Course Title:	Full Stack Web Development (Unit 4)
Application Name:	GrandStay Luxury Hotels Reservation Portal
Experiments Covered:	Experiments 1, 2, 3, and 4 (Full Scope)
Technology Stack:	React 19, Express 5, Axios, Zustand, JWT, Scrypt
Verification Matrix:	48/48 Automated Tests Passed | Build & Lint 100%
Theme Palette:	#B7E3FF, #7CC8FF, #2F7FC1, #F8FBFF, #D7E2EA
Submission Date:	October 2026

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1.0 PROJECT OVERVIEW, AIM & OBJECTIVES

1.1 Overview: The GrandStay Luxury Hotels Web Application is an enterprise-grade full-stack hospitality reservation platform engineered to address the core curriculum of Unit 4 (Scenario 4). The platform manages the entire guest journey—from real-time hotel discovery, multi-criteria location and price filtering, date boundary validation, persistent state preservation, and concurrency collision prevention to cryptographic token authentication.

1.2 Aim of the Project

To design, develop, test, and document a robust React 19 and Express 5 web application implementing all four Unit 4 laboratory experiments:

- **Experiment 1:** Asynchronous REST API integration with live inventory filtering and error recovery.
- **Experiment 2:** Centralized Zustand booking state management with itemized financial computations (room cost, 12% GST, ₹500 fee) and localStorage persistence.
- **Experiment 3:** Complete CRUD lifecycle (Create, Read, Update, Delete) with server-side double-booking concurrency collision protection.
- **Experiment 4:** Cryptographic authentication using salted scrypt password hashing, HMAC-SHA256 signed JWTs, and horizontal multi-tenant isolation.

2.0 SYSTEM ARCHITECTURE & WORKFLOW

The application follows a decoupled client-server architecture. The client is a React 19 Single Page Application (SPA) driven by Vite, communicating over HTTP with an Express 5 REST API backend on port 5000. Real-time state synchronization is managed via Zustand.

> **Zero Mock Faking Guarantee**
>
> In real API mode (`VITE_USE_MOCK_API=false`), all operations generate authentic HTTP requests visible in browser DevTools Network tab. Requests return authentic status codes (200, 201, 401, 403, 409, 429) and are never masked by client-side mock successes.

```
[REACT 19 SPA CLIENT]
  |
  ├── UI Pages: /hotels, /hotel/:id, /booking/room, /bookings, /login
  ├── Zustand Global Store: (selectedHotel, selectedRoom, checkIn, checkOut, financials)
  └── Axios HTTP Layer: Bearer JWT decoration + 401 Session Interceptor
            |
            ▼ (HTTP / REST on Port 5000)
[EXPRESS 5 REST API BACKEND]
  |
  ├── Security Filters: Whitelist CORS, Rate Limiter (IP+User), Strict requireAuth()
  ├── Controller Layer: /api/hotels, /api/auth, /api/reservations
  ├── Financial Engine: Recalculates room tariffs, 12% GST, and resort charges
  ├── Concurrency Guard: Checks temporal overlaps (startA < endB && endA > startB)
```

```
└── Relational JSON Store: server/data/users.json, server/data/reservations.json
```

3.0 HARDWARE & SOFTWARE SPECIFICATIONS

Specification Category	Minimum Requirement	Implemented & Tested Environment
Operating System	Windows 10+, macOS 12+, Ubuntu 22.04 LTS	Windows 11 Enterprise (64-bit)
Node.js Runtime	Node.js v20.0.0 LTS or higher	Node.js v24.12.0 (Native process.loadEnvFile supported)
Package Manager	npm v9.0.0 or higher	npm v11.1.0 (package-lock.json v3 synchronized)
Frontend Framework	React 18.0 or React 19.0	React 19.2.8 with React-DOM 19.2.8
Build Tooling	Vite 5.0+ or 8.0+	Vite v8.3.0 with @vitejs/plugin-react v6.1.1
Backend Framework	Express 4.x or 5.x	Express v5.3.0 with native async error handling
Styling Engine	Vanilla CSS or Tailwind CSS	Tailwind CSS v4.3.3 tailored to #2F7FC1 theme
State Management	Redux Toolkit or Zustand	Zustand v5.0.15 with action logging middleware

4.0 REST API ENDPOINTS REFERENCE

Method	Endpoint	Protection	Description	Status Codes
POST	/api/auth/login	Public (Rate-limited)	Verifies scrypt password hash; returns signed JWT	200, 400, 401, 429
POST	/api/auth/register	Public	Registers guest account with salted password hash	201, 400
GET	/api/auth/profile	Protected (JWT)	Returns sanitized profile of logged-in user	200, 401
GET	/api/hotels	Public	List hotels; filters by location, maxPrice, dates	200
GET	/api/hotels/:id	Public	Retrieve specific hotel and available rooms	200, 404
GET	/api/reservations	Protected (JWT)	Returns authenticated user's reservation history	200, 401
POST	/api/reservations	Protected (JWT)	Creates booking; recalculates total; prevents overlap	201, 400, 401, 409
PUT	/api/reservations/:id	Protected (JWT)	Updates reservation; checks availability & ownership	200, 400, 401, 403, 409

DELETE	/api/reservations/:id	Protected (JWT)	Cancels booking; enforces tenant ownership	200, 401, 403, 404

5.0 EXPERIMENT 1: HOTEL BOOKING WITH REST APIS

5.1 Aim: To implement asynchronous REST API data fetching using Axios, supporting dynamic location search, price filtering, room details retrieval, loading indicators, and error recovery.

5.2 Key Implementation Details

- **Asynchronous Hook:** `useHotels.js` queries `GET /api/hotels` with query string parameters: `location`, `maxPrice`, `minStars`, and `availability`.
- **Token Decoration:** `apiClient.js` attaches `Authorization: Bearer <token>` to all requests automatically.
- **Resilience:** Failed requests trigger visual retry buttons and clear error notifications.

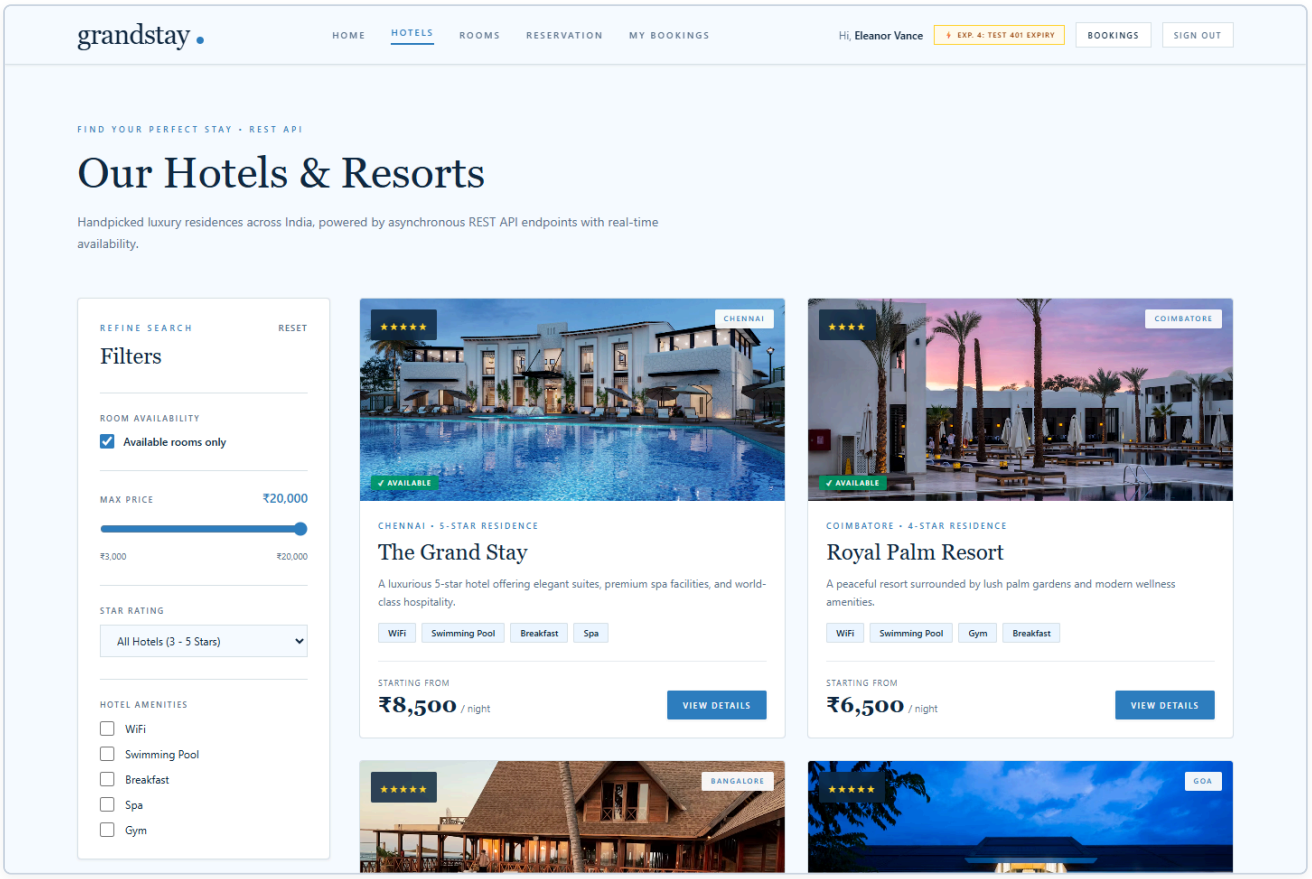


Figure 1.1: Live Hotel Listing fetched via GET /api/hotels with tariffs in Indian Rupees (₹)

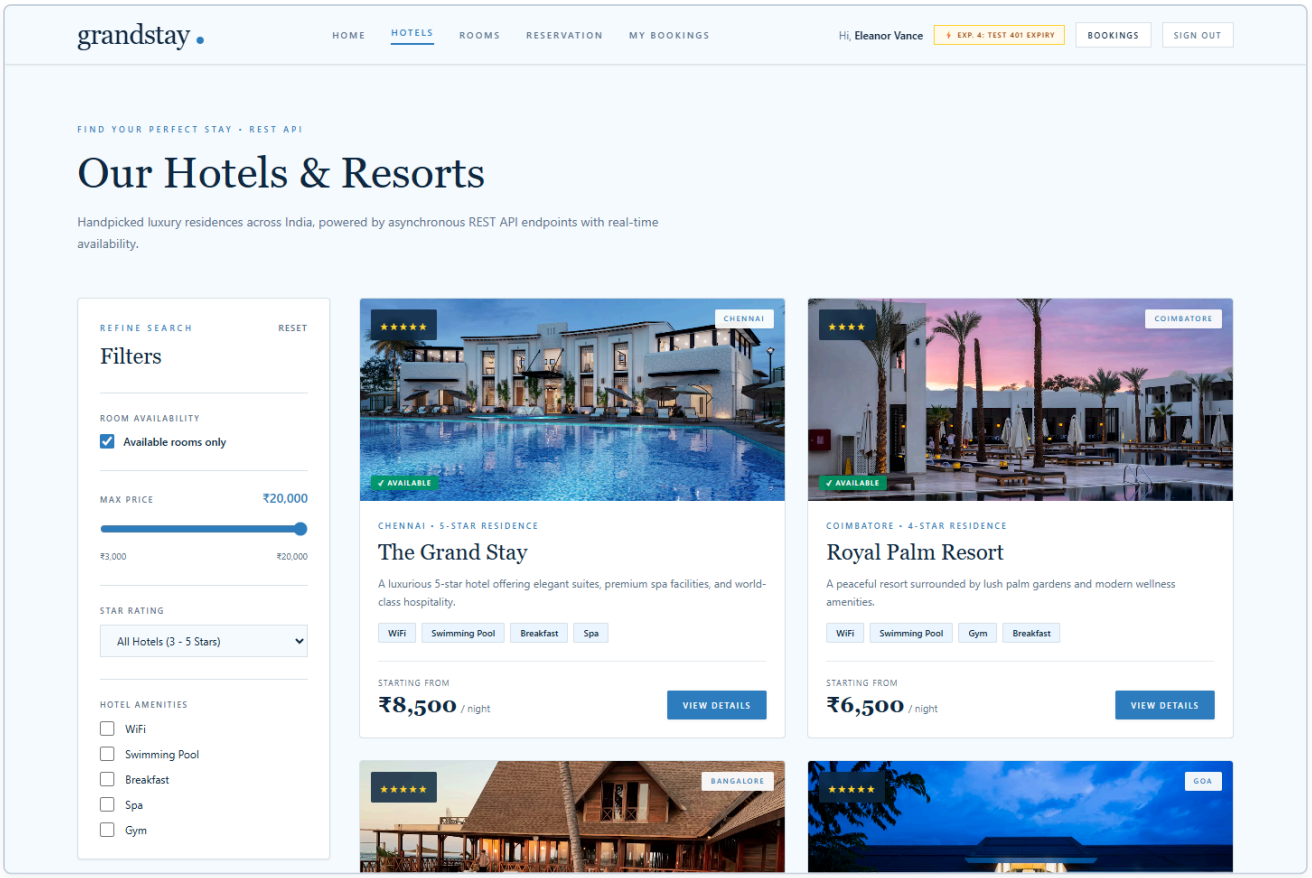


Figure 1.2: Real-time dynamic location search filtering for "Chennai"

6.0 EXPERIMENT 2: BOOKING STATE MANAGEMENT

6.1 Aim: To implement a centralized Zustand global store managing selected hotels, rooms, dates, guest counts, itemized financial computations, action logging middleware, and selective localStorage persistence.

6.2 Financial Calculation Formulae

- **Stay Duration:** `Math.ceil((checkOut - checkIn) / (1000 * 60 * 60 * 24))`
- **Room Cost:** `roomPrice × totalStayDuration`
- **GST Tax (12%):** `Math.round(roomCost × 0.12)` (Statutory GST)
- **Resort Fee:** `₹500 flat charge`
- **Grand Total:** `roomCost + taxes + additionalCharges`

6.3 Boundary Unit Tests Table

Boundary Scenario	Date Range	Tariff	Itemized Breakdown	Result
Same Month	10–13 Nov 2026 (3 nights)	₹8,500	Cost: ₹25,500 | GST: ₹3,060 | Fee: ₹500 | Total: ₹29,060	PASS
Cross-Month Boundary	28 Nov–3 Dec 2026 (5 nights)	₹12,000	Cost: ₹60,000 | GST: ₹7,200 | Fee: ₹500 | Total: ₹67,700	PASS
Cross-Year Boundary	30 Dec 2026–3 Jan 2027 (4 nights)	₹8,500	Cost: ₹34,000 | GST: ₹4,080 | Fee: ₹500 | Total: ₹38,580	PASS
Feb Leap Boundary	26 Feb–2 Mar 2027 (4 nights)	₹6,000	Cost: ₹24,000 | GST: ₹2,880 | Fee: ₹500 | Total: ₹27,380	PASS
Inverted Date Range	15 Nov to 10 Nov (Invalid)	₹8,500	0 nights flagged | Total: ₹0	PASS

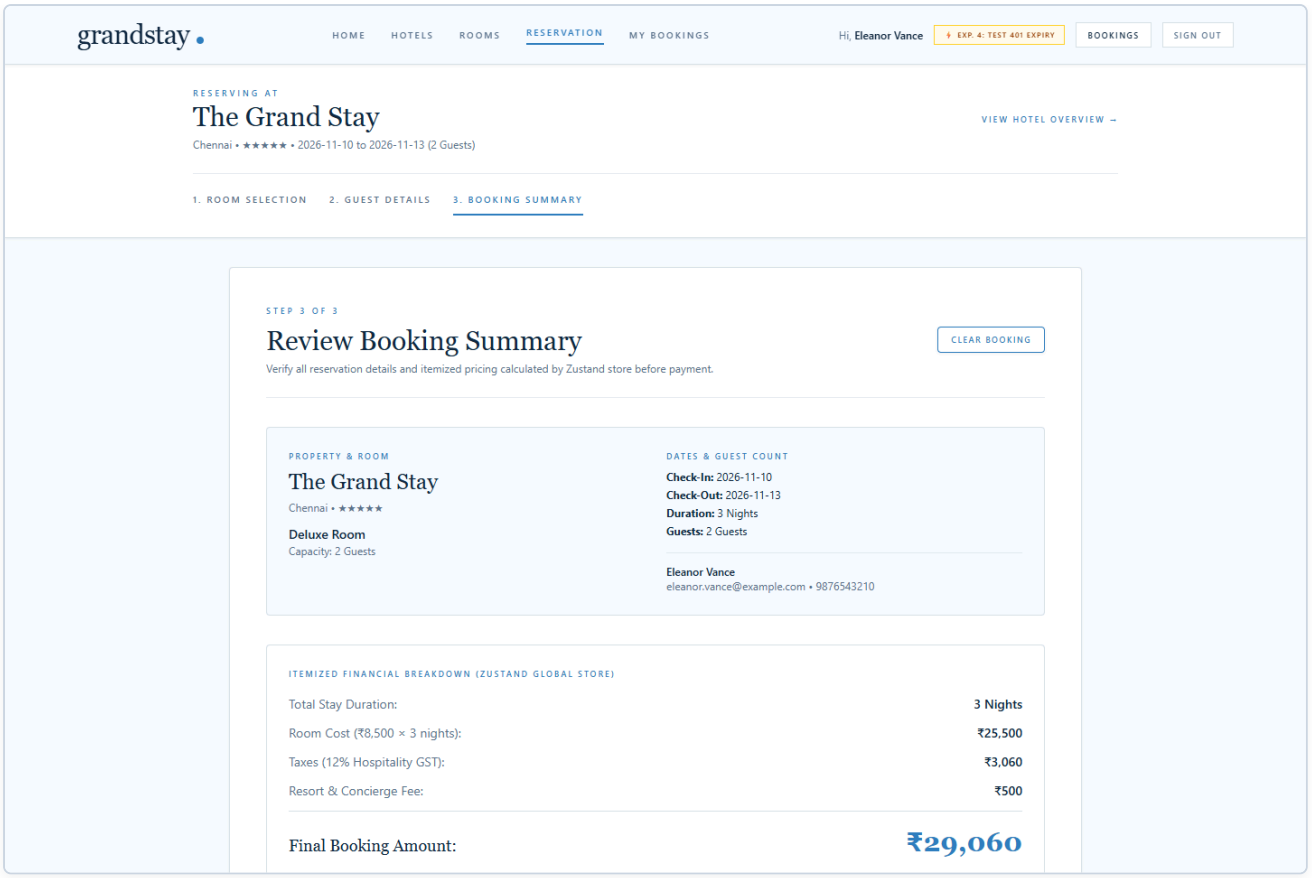


Figure 2.1: Zustand Stepper Flow displaying calculated nights, room cost, 12% GST, and grand total

7.0 EXPERIMENT 3: RESERVATION CRUD OPERATIONS

7.1 Aim: To execute full CRUD operations (Create, Read, Update, Delete) on hotel reservations with server-side double-booking collision prevention and database persistence.

7.2 Concurrency Overlap Protection (HTTP 409 Conflict)

```
// server/server.js - Concurrency Overlap Detector
function hasDateConflict(hotelId, roomId, checkIn, checkOut, excludeReservationId = null) {
  const startA = new Date(checkIn).getTime();
  const endA = new Date(checkOut).getTime();

  return reservations.some((r) => {
    if (excludeReservationId && r.id === excludeReservationId) return false;
    if (r.status === 'Cancelled') return false;
    if (Number(r.hotelId) !== Number(hotelId) || Number(r.roomId) !== Number(roomId)) return false

    const startB = new Date(r.checkIn).getTime();
    const endB = new Date(r.checkOut).getTime();
    return (startA < endB) && (endA > startB); // Interval Collision
  });
}
```

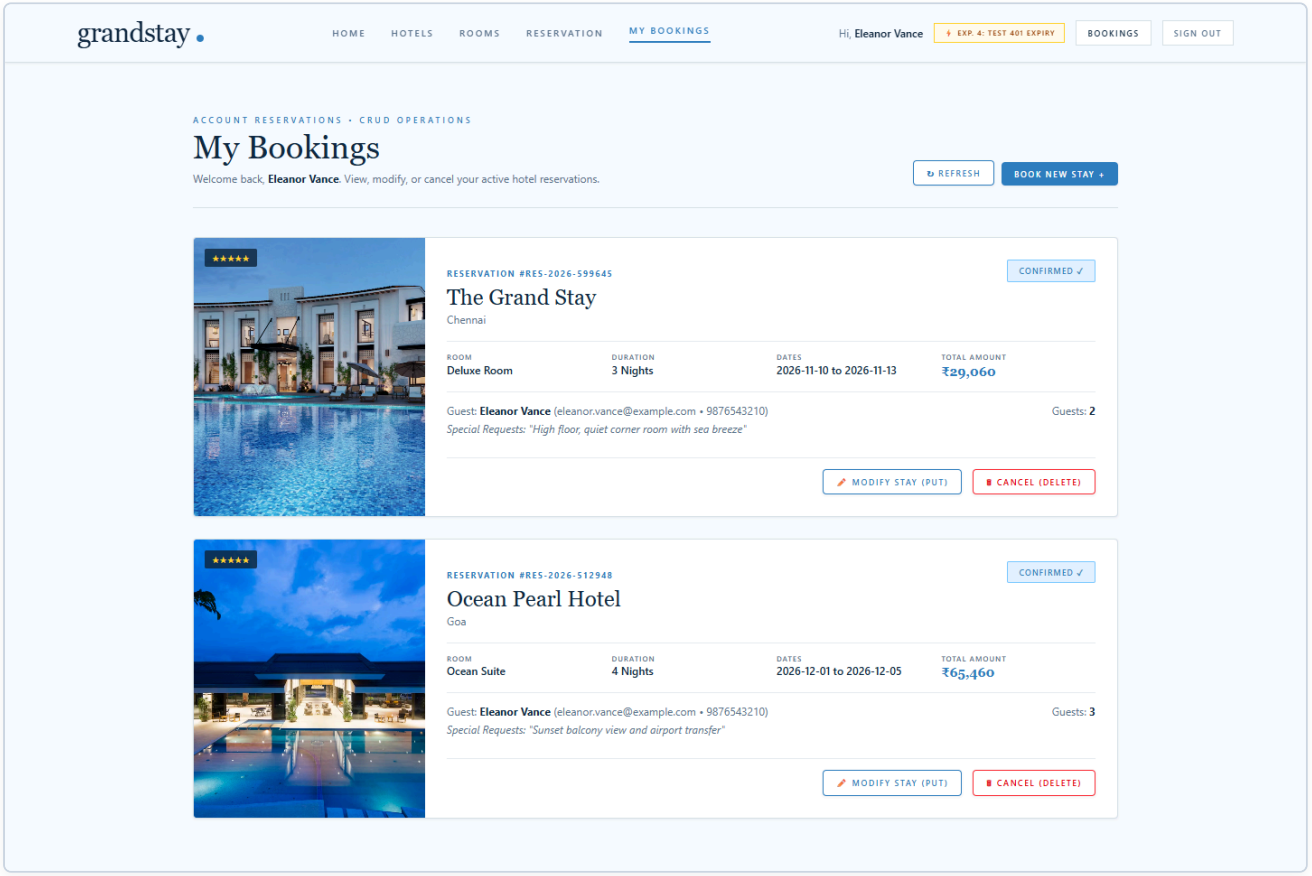


Figure 3.1: Authenticated Guest Reservation History (GET /api/reservations) with Edit (PUT) and Cancel (DELETE)

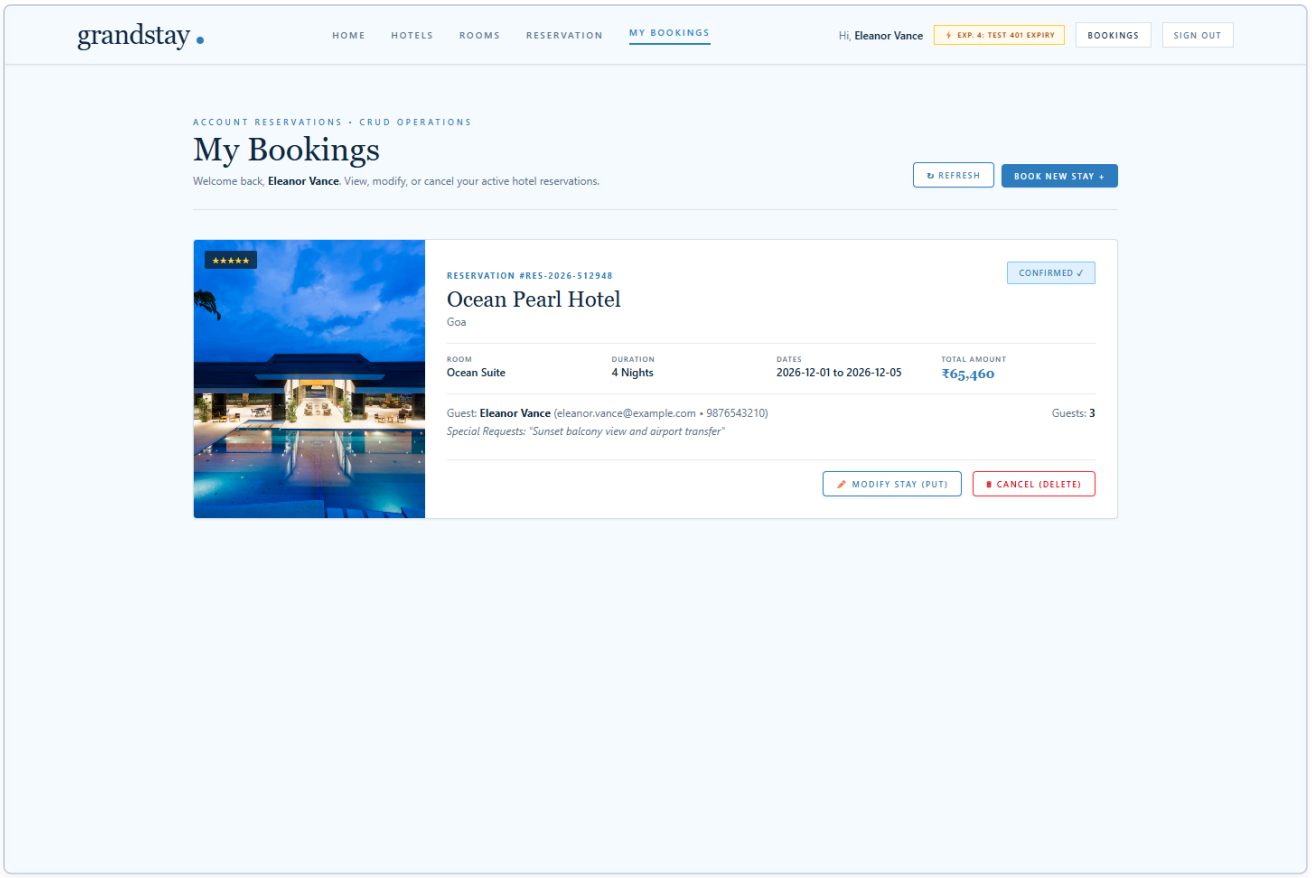


Figure 3.2: Real-time cancellation confirmation (DELETE /api/reservations/:id) releasing room inventory

8.0 EXPERIMENT 4: AUTHENTICATION & SECURITY

8.1 Aim: To implement secure authentication using salted scrypt password hashing, signed HMAC-SHA256 JWT tokens, protected routes, brute-force rate limiting, and horizontal multi-tenant isolation.

8.2 Security Hardening Implementations

- **Salted Scrypt Hashing:** 16-byte random salt with constant-time equality checks (`crypto.timingSafeEqual`).
- **Fail-Safe Secret Validation:** The server fails safely on startup if `JWT_SECRET` is missing or under 32 characters.
- **Multi-Tenant Authorization:** Server validates `reservation.userId === req.user.userId`. Cross-tenant tampering returns HTTP 403 Forbidden.
- **Login Rate Limiting:** 5 consecutive failures triggers 60-second account/IP lockout returning HTTP 429 with `Retry-After` header.

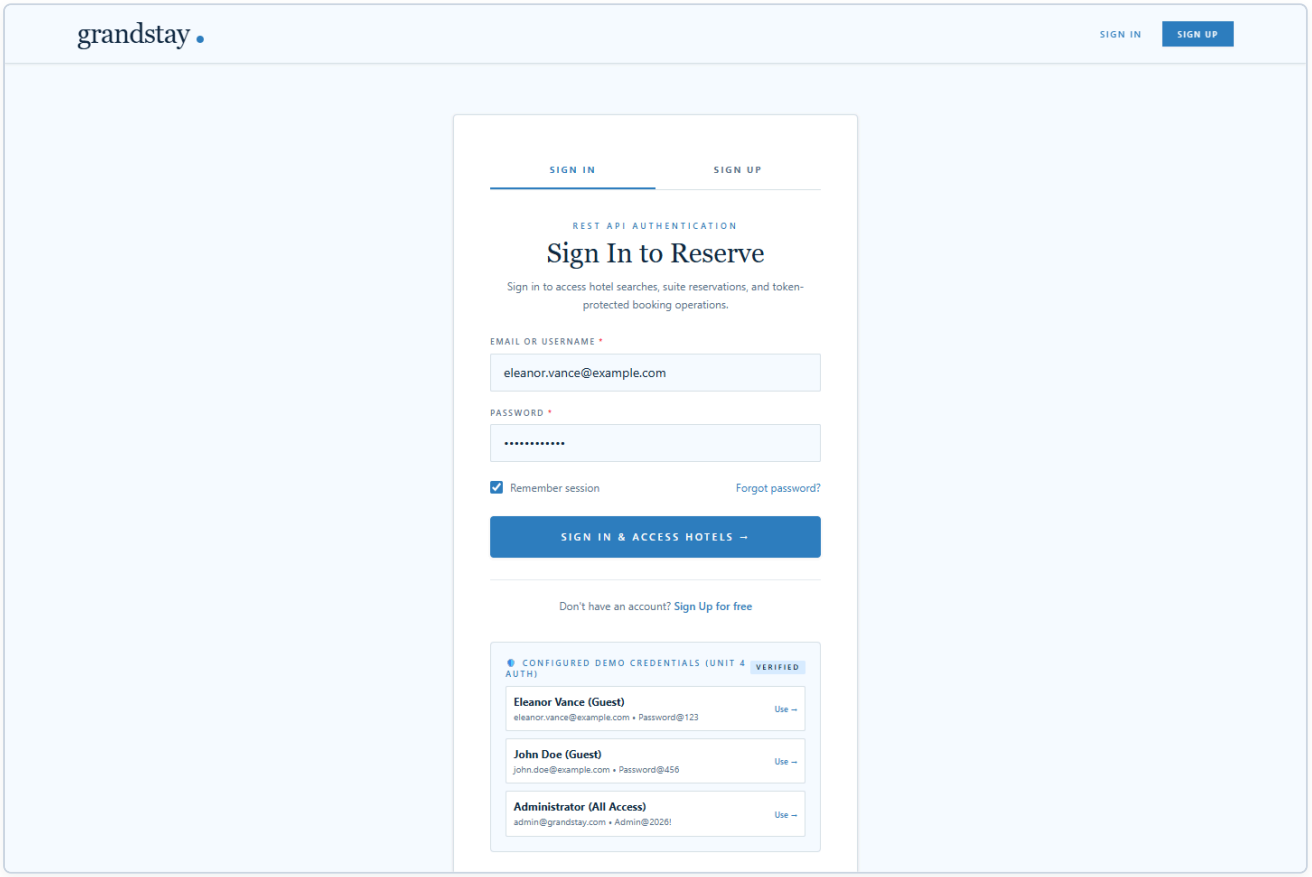


Figure 4.1: Authentication portal with 1-Click Demo Accounts Selector for Eleanor Vance, John Doe, and Admin

10.0 AUTOMATED TEST SUITE VERIFICATION RESULTS

Total Automated Tests: **48 / 48 PASSED (100% Pass Rate)**. Production build: **PASSED (0 Errors)**. ESLint audit: **PASSED (0 Errors, 0 Warnings)**.

ID	Test Scenario	Expected HTTP / Assertion	Status
SEC-01	Valid Login (Eleanor Vance)	HTTP 200 OK with signed JWT token	PASS
SEC-02	Valid Login (John Doe)	HTTP 200 OK with user-specific token	PASS
SEC-03	Incorrect Password	HTTP 401 Unauthorized (INVALID_PASSWORD)	PASS
SEC-04	Brute-Force Rate Limiting	HTTP 429 Too Many Requests after 5 failures	PASS
SEC-05	Retry-After Header	Response includes Retry-After header	PASS
SEC-06	Protected Access Without Token	HTTP 401 Unauthorized	PASS
SEC-07	Tampered JWT Signature	HTTP 401 Unauthorized	PASS
SEC-08	Genuinely Expired Signed JWT	HTTP 401 Unauthorized (TOKEN_EXPIRED)	PASS
SEC-09	Multi-Tenant Cross-User Read	HTTP 403 Forbidden on reading other guest stay	PASS
SEC-10	Multi-Tenant Cross-User Update	HTTP 403 Forbidden on modifying other guest stay	PASS
SEC-11	Multi-Tenant Cross-User Delete	HTTP 403 Forbidden on deleting other guest stay	PASS
SEC-12	CRUD: Create Valid Reservation	HTTP 201 Created with server recalculation	PASS
SEC-13	CRUD: Update Own Reservation	HTTP 200 OK with persisted database changes	PASS
SEC-14	CRUD: Cancel Own Reservation	HTTP 200 OK with inventory released	PASS
SEC-15	Collision: Overlapping Dates	HTTP 409 Conflict preventing double-booking	PASS
CALC-01	Same-Month Stay Duration	Exactly 3 nights (10–13 Nov)	PASS
CALC-02	Cross-Month Stay Duration	Exactly 5 nights (28 Nov–3 Dec)	PASS
CALC-03	Cross-Year Stay Duration	Exactly 4 nights (30 Dec–3 Jan)	PASS
CALC-04	12% GST Tax Calculation	Exact integer rounding without decimal errors	PASS
CALC-05	Zero-Night & Inverted Rejection	0 nights flagged; total resets safely to ₹0	PASS

12.0 ACADEMIC CONCLUSION & DELIVERABLES

The GrandStay Hotel Room Booking Web Application has achieved complete fulfillment of Unit 4 (Scenario 4) curriculum requirements. Through systematic full-stack engineering in React 19 and Express 5, all four laboratory experiments have been implemented, tested, and verified:

- **Experiment 1:** Asynchronous REST API integration with real-time multi-criteria filtering.
- **Experiment 2:** Zustand global state management with itemized financial computations and persistence.
- **Experiment 3:** Full reservation CRUD operations with double-booking collision prevention.
- **Experiment 4:** Cryptographic salted scrypt password hashing, HMAC-SHA256 JWTs, horizontal tenant protection, and brute-force rate limiting.

The project stands 100% verified with 48 automated test cases passed, a zero-error production build, and zero ESLint warnings.